EurekaZero: Supplementary Telemetry

Candidate movement precedes admission; retention and independent learning remain separate questions.

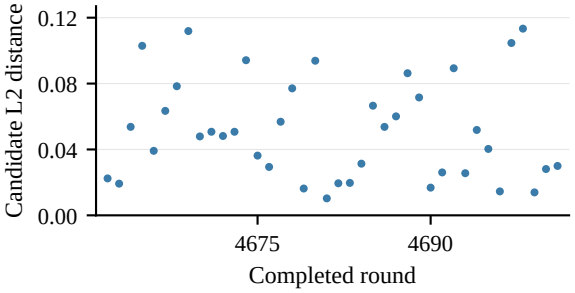


Figure S1: **Recorded candidate displacement varies across rounds.** All 40 complete retained summaries, rounds 4662–4701, captured at 00:55 UTC on 4 November 2026. Points show logged candidate parameter-space L2 displacement; no smoothing or filtering is applied. In the inspected loop, it includes warm-restart perturbation and optimization before admission. The journal records zero admissions in this window. This does not isolate optimizer effects or establish a retained learning gain. Scope: the adaptive adapter admission record in the main manuscript.

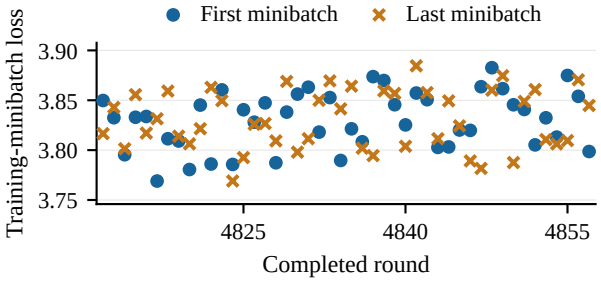


Figure S2: **First and last recorded minibatch losses use different batches.** All 46 complete retained summaries, rounds 4812–4857, captured at 05:53 UTC on 4 November 2026. The 92 points show logged first and last training-minibatch objectives, with two optimizer steps per candidate round. Batch-order hashes differ within every round; these are not matched before/after losses. No smoothing or filtering is applied. The window records zero admissions; these losses establish no retained learning gain. Scope: the adaptive adapter admission record in the main manuscript.

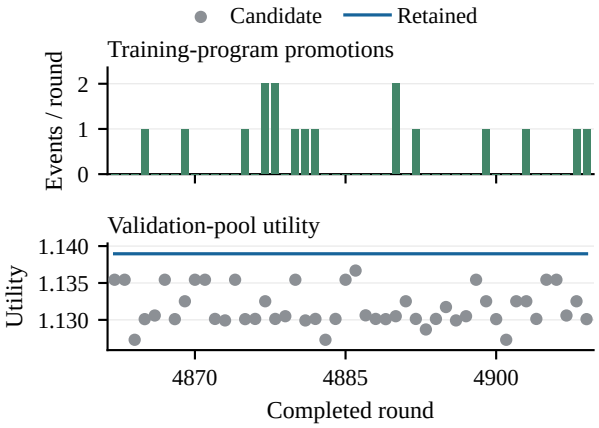


Figure S3: **Program promotion and adapter retention are distinct events.** All 48 complete retained summaries, rounds 4862–4909, captured at 07:21 UTC on 4 November 2026. Top: 17 strict training-program promotions, independently matched to the program journal; zero-event rounds retained. Bottom: candidate and post-decision retained adapter utility (higher is better); zero adapter admissions. Values are unsmoothed, without outcome filtering. Reused development contexts do not establish independent writer or ranking-map learning gains. The inspected caller decides adapter admission before program search; round alignment does not identify a training effect. Scope: the adaptive adapter admission record in the main manuscript.

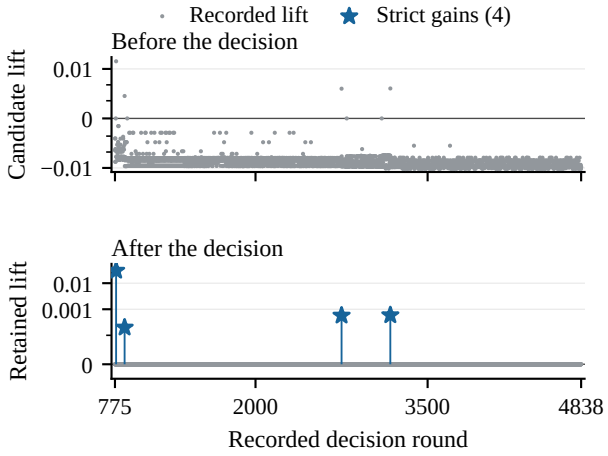


Figure S4: **Logged retained utility increases at four recorded decisions.** All 4,063 admission records in rounds 775–4838, captured at 04:56 UTC on 4 November 2026. Top: candidate utility minus the logged prior retained best; bottom: post-decision retained utility minus that same retained best. Stars mark the four strict gains. The other 4,059 retained differences are zero, including four accepted ties; zero does not imply unchanged weights. Axes are symmetric-log, linear within 10^{-4} utility units of zero. Points retain every observed decision; missing round 4331 is not imputed. These unsmoothed differences on reused development contexts establish neither a causal training effect nor independent writer or ranking-map learning. Scope: the adaptive adapter admission record in the main manuscript.

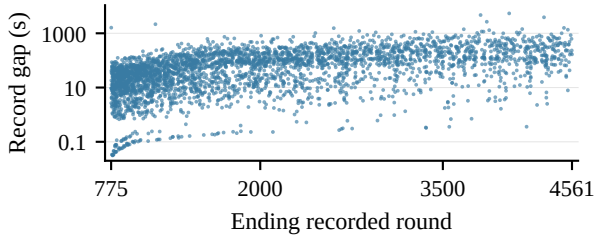


Figure S5: **Program-journal records arrive at varying intervals.** All 3,363 adjacent timestamp gaps (seconds) from 3,364 committed records in rounds 775–4561, captured at 20:57 UTC on 3 November 2026; warm-up records are excluded. Each point is keyed to the later record's round; all pairs are retained. The vertical axis is logarithmic. Recording cadence includes work, waits and batched writes, not per-proposal runtime or adapter admission. These observations establish neither a speed comparison nor learning. Scope: the training-program journal in the main manuscript.

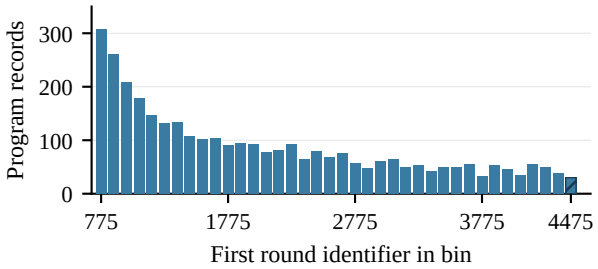


Figure S6: **Program-record counts vary across round windows.** All 3,364 committed program-journal records in rounds 775–4561, captured at 20:57 UTC on 3 November 2026; warm-up records are excluded. Each bar counts records in one fixed 100-round-identifier bin. The hatched final bin spans only 4475–4561 (87 identifiers); its count is not rescaled. Every record enters exactly one bin. Round identifiers do not supply completed-round exposure or a denominator of attempted proposals. These counts establish neither adapter admission nor independent learning. Scope: the training-program journal in the main manuscript.

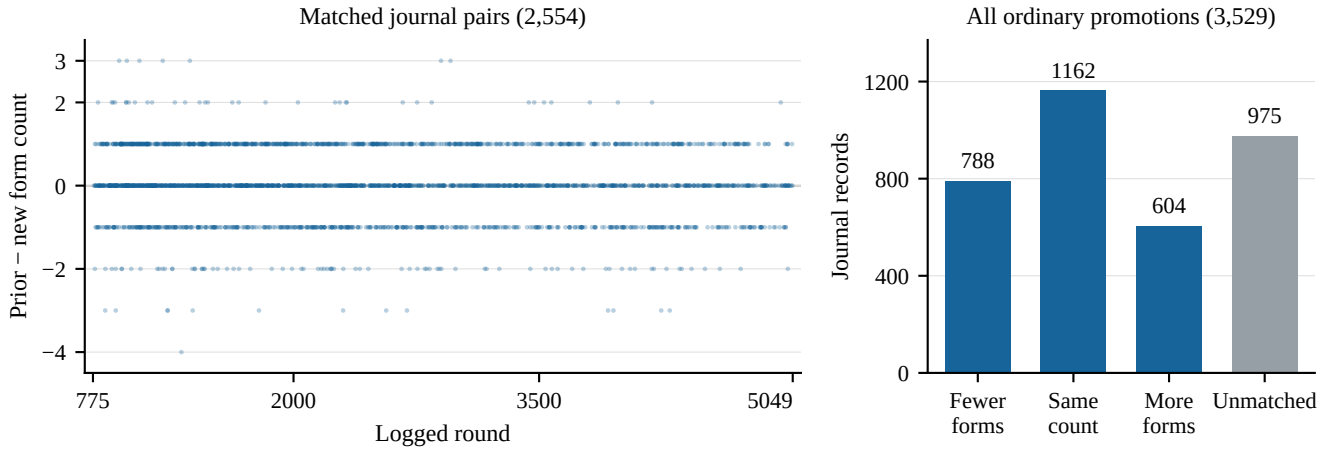


Figure S7: **Matched promotions can simplify or enlarge the counted program.** All 3,529 ordinary program promotions in rounds 775–5049; the last event is at round 5047 (at 11:13 UTC on 4 November 2026). Left: prior minus new opening-parenthesis count for 2,554 records with an exactly matched prior incumbent utility and an unambiguous prior count in the same context's earlier journal records. Positive values mean fewer counted forms. Right: 788 fewer, 1,162 unchanged, 604 more and 975 without a prior match. Unmatched values are omitted from the left panel, not set to zero; every record enters the right panel. No jitter or smoothing is applied. These context-local representation changes are conditional on admission; they do not measure predictive improvement, adapter retention, neural-writer learning or ranking-map learning. Scope: the training-program journal in the main manuscript.

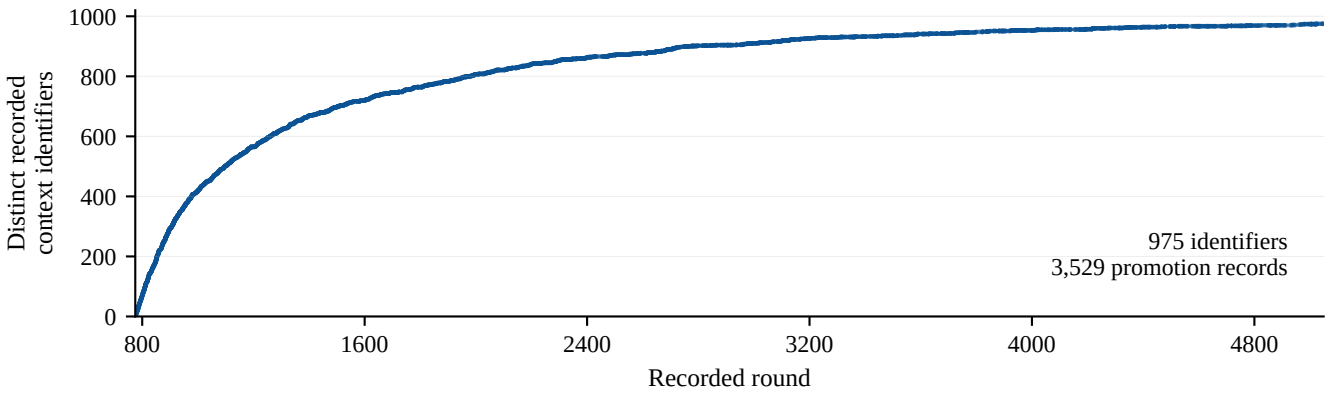


Figure S8: **Recorded promotions carry 975 distinct context identifiers.** All 3,529 ordinary promotion records in rounds 775–5049; the last event is at round 5047 (at 11:13 UTC on 4 November 2026). Each point counts distinct logged context identifiers seen through that record, in source order; repeated identifiers remain in the plot. The count starts at the first record in this window, not the first-ever promotion. No points or connecting lines fill rounds without records. These identifiers do not establish dataset or family coverage or independent tasks. The journal records admitted programs, not all attempted proposals; the count measures neither adapter admission, predictive improvement nor neural-writer or ranking-map learning. Scope: the training-program journal in the main manuscript.

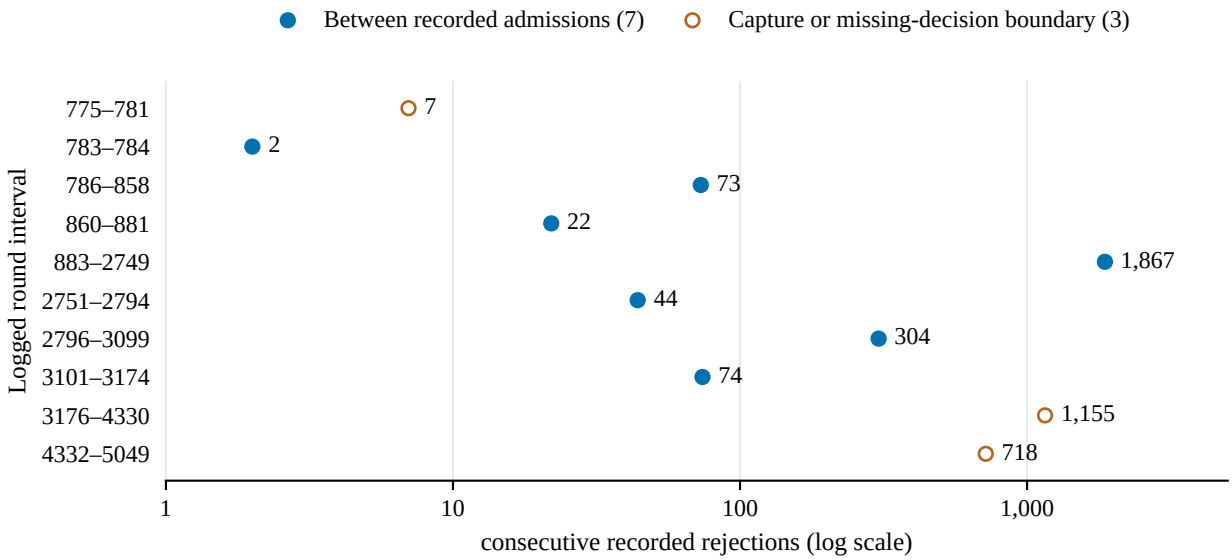


Figure S9: **Recorded adapter rejection runs vary in length.** All 4,274 scored adapter decisions in rounds 775–5049 delimit ten runs containing all 4,266 recorded rejections. Each point counts consecutive rejected round identifiers; labels give their intervals. Filled markers show the seven runs bracketed by recorded admissions; open markers show the three segments touching capture or missing-decision boundaries. Round 4331 has no admission record and breaks, rather than joins, a run. No missing decision is filled. These are adaptive decision counts; independence is not established. They do not measure elapsed training time, learning by a neural writer or ranking map, or progress in program search. Scope: the adaptive adapter admission record in the main manuscript.